

Corticosterone, body condition and locomotor activity: a model for dispersal in screech-owls

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Abstract. A model that explains natal dispersal in resident screech-owls is presented and examined. The model is based on interactions among hormonal changes, body condition and social stimuli. It predicts that corticosterone, an adrenal glucocorticoid known to stimulate locomotor and foraging activity, increases in blood plasma prior to dispersal through a combination of endogenous and exogenous events. This mediates the locomotor activity that underlies dispersal behaviour. Juveniles in good body condition (i.e. those with sufficient fat reserves) will disperse when corticosterone increases. Birds in poor body condition will not, but they will increase their foraging activity under the influence of corticosterone. Dispersal of siblings will reduce aggression and/or competition for food, enabling the remaining juveniles to improve their body condition and disperse. Initial studies on screech-owls, *Otus asio* and *O. kennicottii*, have produced results that are generally consistent with the model. For example, captive juvenile screech-owls showed increased locomotor activity in the weeks leading up to the time when free-living juveniles are dispersing, and activity levels declined thereafter. Peaks in corticosterone corresponded with periods of high locomotor activity (i.e. at the time of dispersal) in captive owls. Finally, field studies indicate that dominant juveniles, which are presumably in better physical condition, initiated dispersal before their more subordinate siblings.

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Dispersal is the movement of individuals from their point of origin (natal area) to where they reproduce or would have reproduced had they survived and mated (Howard 1960). The movement is characteristic of individuals of the juvenile age class (Greenwood 1980), and typically it is permanent; that is, young do not return to natal areas once they have become established following dispersal. Virtually all species of birds make a dispersal movement at some stage of the life cycle and, in many species, females tend to disperse further than males (Gauthreaux 1978; Greenwood 1980). In most cases, however, juveniles of both sexes disperse from parental territories during the post-fledging period, soon after they achieve independence from adults. Because of this, the initial dispersal movements have also been referred to as post-fledging dispersal (e.g. Morton et al. 1991).

Beyond achieving independence from adults, the proximate factors that stimulate young to

initiate dispersal are not well understood. Some factors that have been suggested are parental aggression towards young (Bunn et al. 1982; Wiggett & Boag 1993), aggression of young toward each other (De Laet 1985; Strickland 1991) and declining food availability within the natal area (Veltman 1989; Kenward et al. 1993; King 1996). Despite evidence that such exogenous or environmental factors are important in driving dispersal in some species, they appear less important in others. For example, some birds disperse from natal areas despite the experimental removal of their parents and unlimited access to food (Nilsson 1990). Under these conditions, neither diminishing food supplies nor parental aggression cause young to disperse, and dispersal is probably initiated by the presence of an endogenous mechanism. Finally, relative body condition or social status may account for variations in the timing of dispersal within and among broods; larger and/or more dominant individuals disperse before smaller, subordinate individuals in some species (Nilsson & Smith 1985; Ellsworth 1997). Thus,

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either exogenous or endogenous factors may influence dispersal, or they may interact with each other to result in the final behaviour pattern.

Dispersal probably occurs differently in migratory and sedentary species of birds. In migratory birds, dispersal is temporally divided: one part often occurs in juveniles during the post-fledging period prior to initial autumn migration, and the other part occurs in yearlings soon after they arrive in the vicinity of potential breeding sites following the initial spring migration (Drilling & Thompson 1988; Morton et al. 1991; Morton 1992). Dispersal movements during the post-fledging period in migratory birds may function to facilitate homing, nest-site or territory selection, or to improve body condition of juveniles preparing to migrate (Morton 1992). None the less, the temporal separation of juvenile independence and ultimate territorial occupancy (as a result of the intervening migration to and from wintering areas) alters the dynamics of dispersal in migratory compared to non-migratory species, and often this separation has made dispersal more difficult to understand in migratory species. Commonly, researchers calculate the distance between the natal and first breeding sites for species of migrants, but fewer studies have provided information on when or how such dispersal movements are accomplished. Therefore, it has been difficult to determine whether selection has operated on migratory birds to make the movements that bring them to breeding territories before migration, after migration, or as a result of both combined. In resident species with no pronounced biannual movements, dispersal is not interrupted by migration. Thus, in the absence of the physiological, behavioural and geographic changes associated with migration, it should be easier to identify the important selection pressures currently operating on dispersal, as well as to uncover the important physiological mechanisms driving dispersal in these non-migratory species.

Within non-migratory species, one factor appears relatively clear: selection has usually operated for dispersal to occur early so that it is initiated and completed before winter (Morton 1992). In many species, juveniles make a permanent movement from their natal area in late summer. This movement takes young birds away from parents and their natal habitat, and the young must either settle individually within vacant habitat or join a flock on or near their

subsequent breeding site. These movements may occur over a period of weeks (e.g. Nilsson & Smith 1988; Ekman 1989), but some juveniles settle in a matter of days after departing parental territories (Belthoff & Ritchison 1989). The optimal strategy of dispersing juveniles may be to move only as far as necessary (Murray 1967), although many long-distance dispersers probably pass over unoccupied suitable habitat (e.g. Belthoff & Ritchison 1989).

We have developed a model that attempts to explain the proximate mechanisms that underlie dispersal in two representative non-migratory species of birds that obtain and defend territories in the first year of life: the western screech-owl, *Otus kennicottii*, and its close relative, the eastern screech-owl, *O. asio*. We believe that the model is easily generalized to many non-migratory species of birds and may also have applications to migratory species. The model is based on interactions among body condition, locomotor activity patterns and the adrenocorticoid hormone corticosterone. As such, this new model integrates both exogenous and endogenous factors, and it makes testable predictions concerning the effects of each on dispersal behaviour of juvenile birds. Our objectives in this paper are to (1) introduce and formalize the model and (2) review data from the literature and from initial field and laboratory studies on screech-owls in relation to predictions of the model. Because the study of endocrine mechanisms of dispersal in birds is in its infancy, our hope is that the model and preliminary data will stimulate further experimental work on this subject in a variety of species.

Before presenting the integrated model in detail, we first discuss separately its individual components: body condition, locomotor activity patterns and plasma hormone levels.

BODY CONDITION

Whether the proximate stimulus for natal dispersal is endogenous, exogenous or a combination of both, an organism's metabolic and behavioural response to a particular stimulus may depend, at least in part, on its physical condition at the time the stimulus is applied (e.g. Ketterson & King 1977). For example, in the normal postabsorptive overnight-fasted state, birds do not tap into protein reserves for fuel, because most glucose

production initially occurs via glycogenolysis (e.g. Reisenfeld et al. 1981). However, birds have limited glycogen stores (Blem 1976), and as these decline during food restriction, lipids, the most energy-dense fuel source (Johnston 1970; Cherel et al. 1988), are mobilized and provide an excellent supply of energy (Ramenofsky 1990). Birds have a large capacity to store lipids, and their lipid depots vary on a daily and seasonal basis (Blem 1976, 1990). With prolonged food restriction, however, or with food restriction in birds in poor body condition, gluconeogenesis is enhanced by amino acid metabolism, largely from muscle protein (Siegel 1980; Cherel et al. 1988) and is stimulated by increased glucocorticoid secretion (e.g. Veiga et al. 1978). Furthermore, the extent to which muscle protein is mobilized during fasting is directly related to body fat content at the onset of food restriction and to the size of the energy deficit; lean individuals lose significantly more muscle protein than fat individuals (Cherel et al. 1988). Thus, in relation to dispersal, this suggests that juveniles dispersing in good physical condition would be able to avoid mobilizing muscle protein and would be more likely to survive dispersal. Juveniles without the necessary body condition probably must delay dispersal until their condition improves, through either increased foraging in the natal area or provisioning by adults.

Because early dispersers may have an increased chance of settling in good areas, which presumably enhances survival, selection could operate on young of sedentary species to disperse as soon as they are physically ready (Nilsson & Smith 1985, 1988; Nilsson 1989a, b). For example, early dispersing marsh tits, *Parus palustris*, are heavier than individuals that disperse later, and marsh tits in poorer physical condition forage longer in the natal area than birds in good condition (Nilsson & Smith 1985). As a result, juveniles initially in poor body condition disperse later than juveniles in good condition. In Spanish imperial eagles, *Aquila adalberti*, early dispersers have lower blood urea and uric acid levels than later dispersers (Ferrer 1992, 1993). Because high plasma values of nitrogenous wastes could indicate increased amino acid catabolism (Mori & George 1978; Le Maho et al. 1981) and may reflect individuals in poor condition that are tapping into protein (i.e. muscle) reserves, these data suggest that early dispersers are in better physical condition than late dispersers.

LOCOMOTOR ACTIVITY LEVELS

Increased locomotor activity is probably involved in dispersal (Ritchison et al. 1992; Belthoff & Dufty 1995), and activity levels during dispersal appear to be under endogenous control, at least in part, in the two species of birds examined to date. Ritchison et al. (1992) examined locomotor activity of captive, juvenile eastern screech-owls throughout the pre-dispersal, dispersal, and post-dispersal periods. Despite being isolated from exogenous variables other than photoperiod (e.g. aggression, fluctuating food levels, population pressures), captive owlets increased activity levels in the weeks leading to dispersal of their radio-tagged, free-living counterparts, and activity levels peaked near the time that dispersal occurred in the radio-tagged juveniles. Locomotor activity levels declined thereafter. Similar patterns of locomotor activity levels were detected in captive western screech-owls (Belthoff & Dufty 1995). This dispersal-related activity, which Ritchison et al. (1992) termed 'dispersal restlessness', has offered a means to examine a behaviour pattern in the laboratory that is otherwise very difficult to quantify.

Free-living screech-owls increase locomotor activity near the time of dispersal as well (Ritchison et al. 1992). Similarly, in young mammals there are noticeable behavioural changes as the dispersal period approaches. For example, in Belding's ground squirrels, *Spermophilus beldingi*, the frequency of movement and distance moved per unit time by juvenile males (the dispersing sex) are significantly greater than for females, and these activities peak at the time of dispersal (Holekamp 1986). Juvenile males also spend significantly more time climbing and digging and exploring non-familial burrows and novel objects. It is apparent that increased locomotor activity and novel searching behaviour increase in young animals as they mature, and these probably contribute to the initiation of natal dispersal.

HORMONES

In birds where both sexes of young disperse, they typically do so when gonads are poorly developed. Therefore, it seems unlikely that sex-specific gonadal steroids (e.g. testosterone and oestradiol) control locomotor activity levels and dispersal (but see Holekamp et al. 1984 and Holekamp &

Sherman 1989 for evidence of organizational effects of testosterone on dispersal in young male Belding's ground squirrels). Indeed, our data from western screech-owls indicate that plasma levels of gonadal steroids are low in the weeks and days leading up to dispersal (see below). Instead, we hypothesize that the adrenal glucocorticoid corticosterone is important in the initiation and/or maintenance of high activity levels and dispersal. Corticosterone is present in detectable levels in both sexes during this time period and has been implicated as playing a major role in these phenomena (e.g. Wingfield et al. 1990; Astheimer et al. 1992; Wingfield 1994). Therefore, our model focuses on the actions of corticosterone and its possible role in mediating locomotor activity and dispersal.

Corticosterone is a steroid hormone produced by the adrenal glands that helps to regulate the availability of energy from different sources within the body. The hormone stimulates both anabolic and catabolic processes but is considered to have a net catabolic effect, which is seen primarily in the breakdown of muscle protein to provide energy (Holmes & Phillips 1976). Corticosterone also alters overall body composition by elevating fat levels and reducing lean body mass (Wingfield & Silverin 1986; Bray 1993). Thus, decreases in protein stores are accompanied by increases in body fat.

Corticosterone also has related effects on behaviour; for example, it stimulates increased foraging and food intake (e.g. Nagra et al. 1963). Gearhart et al. (cited in Wingfield 1988) observed that birds given exogenous corticosterone spent more time foraging than birds given metyrapone, a drug that inhibits corticosterone synthesis. This effect is enhanced if birds are food-restricted (Astheimer et al. 1992). The latter result is comparable to earlier findings (Klandorf & Harvey 1984), where adrenocortical function increased in fasted birds and decreased when birds were re-fed. Similarly, activity levels increase in undernourished individuals (e.g. Klandorf & Harvey 1984; Astheimer et al. 1992; Piersma & Poot 1993), a physiological state that stimulates glucocorticoid secretion (Cherel et al. 1988). Wingfield (1994) argued that corticosterone is associated with various types of dispersal-related behaviour patterns, ranging from irruptive migratory events to altered foraging patterns. He, too, suggested that body condition affects the type of movement activity

pursued. Moreover, emigrating juvenile willow tits, *P. montanus*, significantly increased corticosterone levels during the autumn migration period (Silverin 1979; Silverin et al. 1989), and Silverin et al. (1989, page 148) suggested that 'high corticosterone levels are involved in the emigration of juveniles out of the coniferous forest'. Finally Silverin (1997) found that willow tits given subcutaneous corticosterone during the normal dispersal period (late summer/early autumn) disappeared (presumably dispersed) from the study area more frequently than control birds, which received no exogenous hormone, although the fate of the individuals was unknown. Given amassing evidence of its effects on locomotor activity and movement patterns in birds, it is reasonable that a model of natal dispersal incorporate this hormone and its actions.

STUDY SPECIES

Our model is based upon the biology of two representative species of non-migratory birds: eastern and western screech-owls. These owls are common to eastern and western North America, respectively, and sufficient data are now available from both field and laboratory studies to begin to understand factors affecting their dispersal movements. Eastern screech-owls occur east of the Rocky Mountains, where they are permanent residents of both rural and urban habitats from south of the Canadian boreal forest to near the Tropic of Cancer in Mexico (Gehlbach 1995). Adults are primarily socially monogamous and occupy territories throughout the entire year (Ritchison et al. 1988; Belthoff et al. 1993). The post-fledging behaviour and dispersal of radio-tagged eastern screech-owls in Kentucky was described by Belthoff & Ritchison (1989, 1990), Ritchison et al. (1992), Belthoff et al. (1993) and Sparks et al. (1994), from which we generated the following summary. Briefly, young eastern screech-owls fledge in mid-May and spend roughly the next 5 weeks in close association with adults. After this time, young owls increase their independence from adults by roosting further away, increasing the sizes of their home ranges and wandering outside the nightly ranges of their parents more frequently. During this same period, locomotor activity levels of both captive and free-living owls are high. Young owls of both

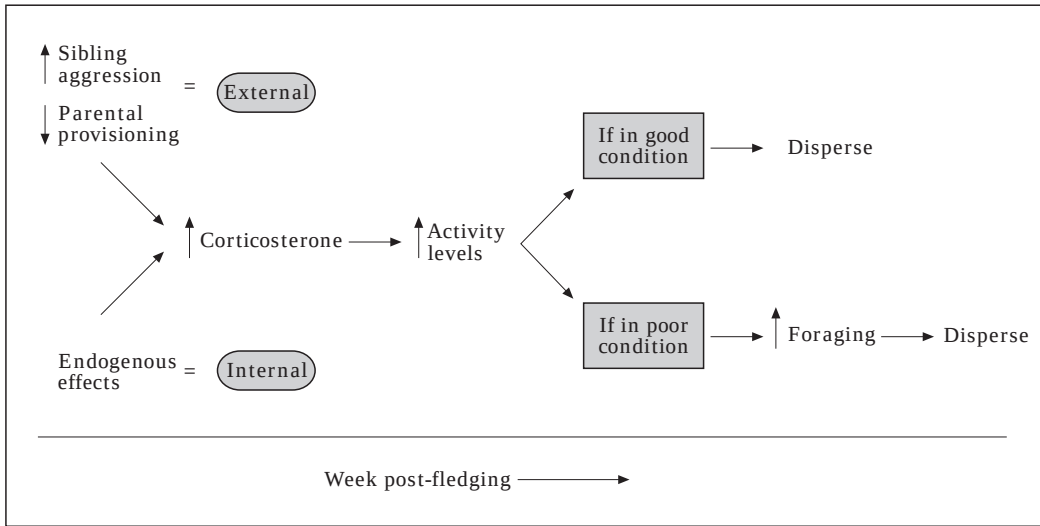


Figure 1. Hypothesized relationship between corticosterone, locomotor activity, body condition and dispersal in screech-owls and other similar non-migratory birds.

sexes initiate dispersal approximately 8 weeks after fledging, usually in mid-July, although the timing of dispersal within and among broods varies. For example, 0–9 days separated the dispersal of the first and last members of broods. Once a young owl initiates dispersal, rather than backtracking frequently or following a more elaborate search pattern, it tends to travel in the same compass direction until it settles to overwinter. Many of the dispersing owls settle on overwinter sites that are 1–17 km from the natal site. Although the number of birds was small, individuals settled approximately 6 days, on average, after initiating dispersal, and some individuals bred in or near these overwinter sites.

Western screech-owls are also non-migratory, socially monogamous and territorial throughout the year (Herting & Belthoff, *in press*), but they occur primarily west of the Rocky Mountains. They are common in low elevation forests, riparian woodlands and suburban areas (personal observation). Most aspects of the post-fledging and dispersal behaviour of western screech-owls are similar to those of their eastern congener (Belthoff & Dufty 1995; Ellsworth & Belthoff, *in press*; unpublished data). In our study area in Idaho, for example, young fledge from nests in mid-May to mid-June and spend approximately 8 weeks on natal territories after fledging and prior to initiating dispersal (personal observation). Both

sexes of young disperse from the natal area, but females ($\bar{X} \pm \text{SE} = 15.8 \pm 3.0$ km, $N = 15$) travel further than males (5.5 ± 1.3 km, $N = 16$; Ellsworth & Belthoff, *in press*) in the first 12 weeks following dispersal. Captive, juvenile western screech-owls also show increased locomotor activity near the time period when free-living juveniles are dispersing (Belthoff & Dufty 1995). Because of similarities between the two species, and because suitable data were available from each species to develop the important components, we believe that our dispersal model is applicable to both species of screech-owls, as well as to other similar species of birds.

CORTICOSTERONE, BODY CONDITION AND LOCOMOTOR ACTIVITY LEVELS: A MODEL FOR DISPERSAL

We hypothesize that screech-owl dispersal is influenced by plasma corticosterone (Fig. 1). As the post-fledging period progresses, parents reduce their provisioning of young and/or aggression increases among siblings. Aggression may increase because of competition for attention of parents or for other resources in short supply in the natal area. Any resulting reduction in food intake by juveniles may cause stress and increase adrenal

activity, leading to increased secretion of corticosterone. Increased plasma corticosterone levels stimulate or are accompanied by increased foraging behaviour and/or locomotor activity.

We also hypothesize that corticosterone levels interact with body condition to induce natal dispersal. When corticosterone secretions increase, activity levels increase, and this eventually leads young to disperse. However, only juveniles in good physical condition are prepared for immediate dispersal. That is, only juveniles with the necessary fat stores will survive the rigours of territory establishment, territory defence and independent foraging. Good physical condition includes adequate lipid reserves, which provide a buffer against possible corticosterone-induced depletion of muscle tissue. Thus, juveniles are protected from the potentially debilitating effects of protein catabolism at a time when functional muscle activity is essential for survival during dispersal activity (see [Holberton et al. 1996](#) for similar arguments concerning corticosterone and migration in birds). On the other hand, juveniles in poor condition at the time that plasma corticosterone levels increase will be stimulated to forage more in the natal area to attain the physical condition necessary to disperse successfully. The model predicts that corticosterone levels remain elevated in these birds, stimulate increased foraging behaviour that increases fat deposition and, eventually, promotes dispersal. That is, following dispersal of their siblings in good condition, the remaining juveniles undergo reduced competition for food, experience less aggression from siblings, and have the exclusive attention of the parents. This improves their physical condition and eventually leads to dispersal by these juveniles as well. Thus, the model predicts a relationship between physical condition, the extent to which corticosterone levels remain elevated, and the timing of dispersal, such that birds in good physical condition leave the natal area first after experiencing only a short period of elevated plasma corticosterone ([Fig. 1](#)). Later dispersers, therefore, are exposed to elevated corticosterone longer than earlier dispersers which may lead to more pronounced catabolic effects on muscle protein. One potential consequence may be reduced survival for later dispersers if they are exposed to chronically elevated corticosterone for too long. Finally, gender differences either in the production of or response to increased plasma corticosterone levels

may play a role in generating the sex-related variation in dispersal distances that have been observed in many avian species.

A variant of the above model holds that the corticosterone increase that stimulates locomotor activity and natal dispersal is body condition-dependent. That is, only birds in good condition have elevated plasma corticosterone levels and disperse. Birds in poor condition maintain lower levels of corticosterone until their body condition improves to a threshold level, at which point they, too, disperse. This construct differs from that shown in [Fig. 1](#) in that only birds in sufficiently good condition begin the increased corticosterone→increased locomotor activity levels→dispersal pathway. Similarly, [Holekamp and her colleagues](#) (reviewed in [Holekamp & Sherman 1989](#)) explained male dispersal in mammals based, in part, on achievement of a threshold body mass. However, the present model is different in that it proposes an activational role for corticosterone. The central tenets of these and other possible variants of this model are the interactions among corticosterone, locomotor activity and body condition. Each form makes testable predictions, and empirical data will eventually enable us to distinguish among them.

Increases in pre-dispersal locomotor activity may be under endogenous control (possibly involving corticosterone), as suggested by both the eastern and western screech-owl locomotor activity data ([Ritchison et al. 1992](#); [Belthoff & Dufty 1995](#)). That is, exogenous stimuli related to the termination of parental feeding, increased aggression among siblings or other factors may not be required for locomotor activity levels to increase, although they may provide supplemental cues that affect the precise timing or amplitude of the corticosterone increase. If so, then elevated corticosterone levels are expected in captive juveniles at the approximate time of dispersal in free-living birds, even when the former are fed continually and isolated from the effects of social interactions.

Finally, gonadal steroids, specifically testosterone, have organizational effects on dispersal behaviour in some small mammals ([Holekamp et al. 1984](#)). It is possible that early organizational effects of sex steroids may establish differences in male and female birds in their hypothalamo-pituitary-adrenal axis that could give rise to sex-biased dispersal patterns. The present model

concentrates on activational mechanisms, however, and does not address organizational processes.

INITIAL DATA FROM SCREECH-OWLS IN RELATION TO THE MODEL

Locomotor Activity Levels

The model predicts that activity levels increase in the weeks leading up to dispersal and decrease thereafter when dispersal is completed. Because the model was conceptualized based on data in [Ritchison et al. \(1992\)](#), examination of locomotor activity levels in relation to dispersal in western screech-owls ([Belthoff & Dufty 1995](#)) provided an additional test of this prediction.

Using birds isolated from exogenous factors other than photoperiod since early in life, [Belthoff & Dufty \(1995\)](#) performed laboratory studies to determine whether increased locomotor activity, triggered by endogenous factors, occurs in young western screech-owls near the time of dispersal. Young owls were individually housed in wire-mesh cages placed in sound-attenuated isolation chambers. Once in chambers, owls could not interact visually or vocally. Owls were provided laboratory mice, day-old chickens and water, and were kept on natural photoperiods. Blood samples were collected from the alar vein once per week to determine circulating levels of hormones ([Belthoff & Dufty 1995](#); [Dufty & Belthoff 1997](#)).

Using digital pedometers, locomotor activity levels were measured for approximately 14 weeks post-fledging. Captive western screech-owls increased locomotor activity in the weeks leading to dispersal and exhibited high activity during the period when dispersal occurs in free-living owls (figure 1 in [Belthoff & Dufty 1995](#)). Activity levels typically declined steadily following this period. Relatively high activity levels before and during the predicted dispersal period suggest that, as in eastern screech-owls, natal dispersal in western screech-owls has an endogenous component that influences locomotor activity levels at the biologically appropriate time.

Body Condition

The model predicts that individuals in adequate body condition will disperse when corticosterone

levels increase. Individuals in poorer body condition will continue to forage on the natal area and initiate dispersal later. There have been no direct tests of the effects of body condition on the dispersal behaviour of screech-owls. However, [Ellsworth \(1997\)](#) examined effects of social dominance on the timing of dispersal in western screech-owls inhabiting the Snake River canyon in southwestern Idaho. This study used video-cameras to record dominance interactions among nestlings, and followed young with radio-telemetry during the post-fledging period and as they dispersed. Individuals scored as dominant were the first to disperse in six of seven (83.3%) broods for which complete information was available. Moreover, the most subordinate juvenile dispersed last in five of seven broods. In four broods, the order of dispersal among juveniles (as many as five young in a brood) correlated perfectly with their relative dominance status within the brood. If dominance status is a predictor of body condition in juvenile western screech-owls (i.e. assuming that dominant birds have the best body condition), then these data provide at least indirect support for the body condition prediction of the dispersal model.

Because the captive, juvenile eastern and western screech-owls examined by [Ritchison et al. \(1992\)](#) and [Belthoff & Dufty \(1995\)](#), respectively, received food ad libitum, there is also no direct information on the effects of body condition on locomotor activity levels (i.e. dispersal restlessness) of owls in captivity. The model predicts that increases in corticosterone should be correlated with increased dispersal activity in individuals with good body condition, and with increased foraging activity in birds in poorer body condition. Studies that manipulate body condition will be required to test these predictions adequately.

Hormones

Both sexes of young screech-owls (eastern and western species) disperse from natal areas in mid-summer ([Belthoff & Ritchison 1989](#); [Belthoff & Dufty 1995](#)). Thus, unlike in many species of mammals, where individuals of one sex class remain on natal areas while individuals of the other sex class disperse (e.g. [Pusey 1987](#)), most (if not all) screech-owls make a dispersal movement. This movement occurs under long but decreasing photoperiods that are non-stimulatory to the

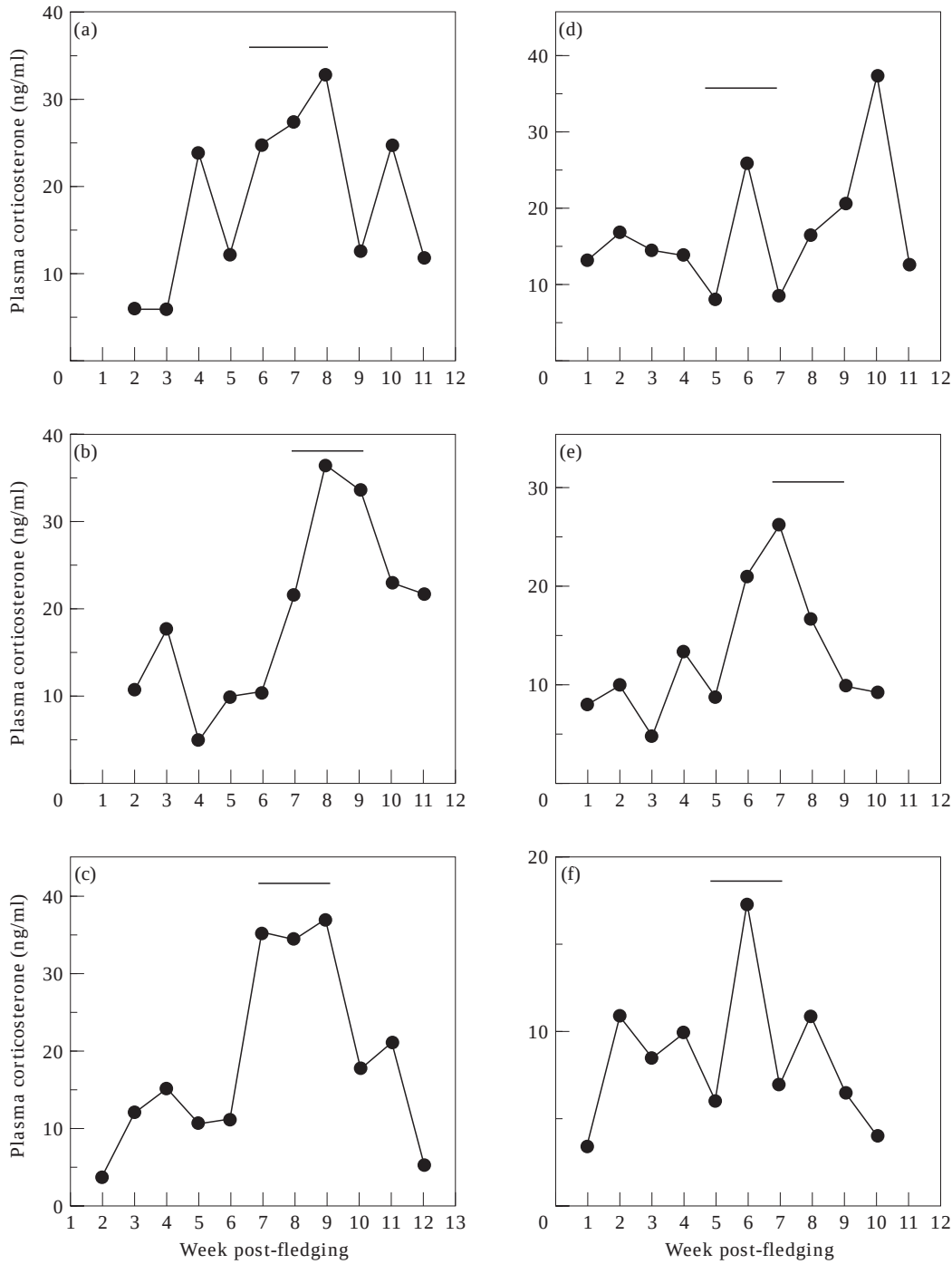


Figure 2. Plasma corticosterone in six captive western screech-owls during the post-fledging period. The bar in each figure represents the 3-week period during which time peak locomotor activity levels were exhibited as measured by back-pack pedometers. Free-living western screech-owls disperse an average of 8 weeks post-fledging.

refractory hypothalamo-pituitary-gonadal axis of juvenile birds (Follett 1991; McNaughton et al. 1992a, b). Therefore, it seems unlikely that the principal gonadal steroid hormones, testosterone and oestradiol, would have activational effects on natal dispersal, and the model predicts that circulating gonadal steroids will either be low in concentration or will not affect dispersal. Our initial studies provide evidence for the former. Using mist nets placed in natal areas, we recaptured 10 free-living, juvenile western screech-owls ($N=6$ males and 4 females) after fledging during summer 1994 and 1995 between 2 and 58 days ($\bar{X} \pm \text{SE} = 25.7 \pm 6.8$ days) before they dispersed. All 10 young owls had testosterone and oestradiol levels that were below the detection level of our radioimmunoassay (see Belthoff & Dufty 1995; Dufty & Belthoff 1997 for methods of blood collection and radioimmunoassay of plasma steroid hormones). Given such low circulating levels, it seems unlikely that these gonadal steroids have activational effects on dispersal in screech-owls.

Corticosterone in captive owls

Circulating corticosterone levels were elevated in six captive owls before or during the time when dispersal restlessness occurred (Fig. 2). Plasma corticosterone levels (assayed from blood samples obtained 0–3 min after birds were removed from their home cage) generally increased during the pre-dispersed period and, in five of six birds, peaked 1–2 weeks prior to the average date of natal dispersal (i.e. 8 weeks after fledging) in free-living owls. These data point to a relationship between corticosterone and locomotor activity during dispersal restlessness, but currently they do not allow us to determine causality. Experiments that manipulate either one or both of these variables are required before the causal relationships are understood. None the less, the data suggest that corticosterone may be an important mediator of the dispersal restlessness behaviour reported in both eastern and western screech-owls.

CONCLUSIONS

We have outlined a model that examines the interplay between endogenous and exogenous factors that regulate natal dispersal behaviour in both eastern and western screech-owls. Initial data on the relationships among corticosterone,

locomotor activity and body condition are generally in agreement with the model. However, the model remains to be tested fully. For example, cause-effect relationships between corticosterone and locomotor activity have not been established. Although the model will undoubtedly be refined with future work, we believe that it is an important first step in understanding the physiological ecology of dispersal in screech-owls and other non-migratory species of birds, and it may provide a useful construct within which dispersal in migratory species could be examined. Confirmatory experiments that manipulate body condition, plasma corticosterone and activity levels are planned as the next step in testing the model and in understanding more fully the physiological ecology of dispersal behaviour in birds.

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